

JEE-Main-06-04-2026 (Memory Based)

[MORNING SHIFT]

Physics

Question: In AC series circuit supply voltage $V_{\text{rms}} = 100$ volts; $R = 80\Omega$; $X_L = 80\Omega$ and source frequency is $f = 50$ Hz. Find the power factor

Options:

- (a) $1/\sqrt{2}$
- (b) $1/2$
- (c) $3/4$
- (d) $\sqrt{3}/2$

Answer: (a)

Solution:

1. Identify the Given Values

From the image, we have:

- Resistance (R): 80Ω
- Inductive Reactance (X_L): 80Ω
- Supply Voltage (V_{rms}): 100 V
- Frequency (f): 50 Hz

2. Calculate Total Impedance (Z)

In a series RL circuit, the total impedance is the vector sum of the resistance and the reactance:

$$Z = \sqrt{R^2 + X_L^2}$$

Plugging in our values:

$$Z = \sqrt{80^2 + 80^2}$$

$$Z = \sqrt{6400 + 6400}$$

$$Z = \sqrt{2 \times 6400}$$

$$Z = 80\sqrt{2}\Omega$$

3. Calculate the Power Factor ($\cos \phi$)

The power factor is defined as the ratio of the resistance to the total impedance:

$$\text{Power Factor} = \cos \phi = \frac{R}{Z}$$

Substituting the values we found:

$$\cos \phi = \frac{80}{80\sqrt{2}}$$

$$\cos \phi = \frac{1}{\sqrt{2}}$$

Question: An expression of potential energy $= \frac{A\sqrt{x}}{B+x}$ is given. Then dimensions of (A.B) will be (x is positive here.)

Options:

- (a) MLT^{-1}
- (b) $M^{1/2}L^{2/3}T^{-2}$
- (c) $ML^{3.5}T^{-2}$
- (d) $ML^{1/2}T^{-2}$

Answer: (3)

Solution:

The given expression is:

$$U = \frac{A\sqrt{x}}{B + x}$$

1. Find the dimensions of B

In the denominator, B is added to x . Since x represents position or distance, its dimension is $[L]$.

Therefore, B must have the same dimension:

$$[B] = [L]$$

2. Find the dimensions of A

Now we look at the entire equation for potential energy (U). The dimensions of energy are $[ML^2T^{-2}]$.

Substituting the known dimensions into the formula:

$$[ML^2T^{-2}] = \frac{[A] \cdot [L]^{1/2}}{[L] + [L]}$$

Since $[L] + [L]$ results in a dimension of $[L]$, we have:

$$[ML^2T^{-2}] = \frac{[A] \cdot [L]^{1/2}}{[L]}$$

$$[ML^2T^{-2}] = [A] \cdot [L]^{-1/2}$$

Isolating $[A]$:

$$[A] = [ML^2T^{-2}] \cdot [L]^{1/2}$$

$$[A] = [ML^{2.5}T^{-2}]$$

3. Find the dimensions of $(A \cdot B)$

Multiply the dimensions of A and B :

$$[A \cdot B] = [ML^{2.5}T^{-2}] \cdot [L]$$

$$[A \cdot B] = [ML^{3.5}T^{-2}]$$

Final Answer:

Comparing this to the options:

- $A: MLT^{-1}$
- $B: M^{1/2}L^{2/3}T^{-2}$
- $C: ML^{3.5}T^{-2}$
- $D: ML^{1/2}T^{-2}$

The correct option is (C).

Question: In YDSE experiment a glass slab of thickness $8\mu\text{m}$ is introduced in front of a slit. If the central maxima shifts to a position of 4th minima, then find refractive index of glass slab

(take $\lambda = 500 \text{ nm}$).

Options:

- (a) 1.22
- (b) 1
- (c) 2
- (d) 0

Answer: (a)

Solution:

1. Identify the Given Values

- Thickness of slab (t): $8\ \mu\text{m} = 8 \times 10^{-6}\ \text{m}$
- Wavelength (λ): $500\ \text{nm} = 500 \times 10^{-9}\ \text{m} = 0.5 \times 10^{-6}\ \text{m}$
- Shift position: The central maxima shifts to the position of the 4th minima.

2. Formulas for Shift and Minima

The linear shift (Δy) produced by a glass slab of refractive index μ and thickness t is:

$$\Delta y = \frac{D}{d}(\mu - 1)t$$

The position of the n^{th} minima from the center in YDSE is given by:

$$y_n = (2n - 1) \frac{\lambda D}{2d}$$



For the 4th minima ($n = 4$):

$$y_4 = (2 \times 4 - 1) \frac{\lambda D}{2d} = \frac{7\lambda D}{2d} = 3.5 \frac{\lambda D}{d}$$

3. Equate and Solve for μ

Since the central maxima shifts to the 4th minima, we set $\Delta y = y_4$:

$$\frac{D}{d} (\mu - 1)t = 3.5 \frac{\lambda D}{d}$$

Cancel out the common terms (D/d):

$$(\mu - 1)t = 3.5\lambda$$

Now, substitute the values for t and λ :

$$(\mu - 1)(8 \times 10^{-6}) = 3.5(0.5 \times 10^{-6})$$

$$(\mu - 1) \times 8 = 1.75$$

$$\mu - 1 = \frac{1.75}{8}$$

$$\mu - 1 = 0.21875$$

$$\mu = 1.21875$$

Rounding to two decimal places, we get $\mu \approx 1.22$.

Question: A block of mass m is released from height h on smooth plane. If normal force on top of the circular part is $3mg$ Find h .



Options:

- (a) $5R$
- (b) $4R$

(c) $3.5R$

(d) $3R$

Answer: (b)

Solution:

1. Forces at the Top of the Circular Path

When the block is at the top of the loop (at height $2R$ from the ground), two forces act downward toward the center of the circle:

1. Weight (mg)
2. Normal Force (N)

The sum of these forces provides the required centripetal force:

$$N + mg = \frac{mv^2}{R}$$

Given that the normal force $N = 3mg$:

$$3mg + mg = \frac{mv^2}{R}$$

$$4mg = \frac{mv^2}{R}$$

$$v^2 = 4gR \quad \text{--- (Equation 1)}$$

2. Conservation of Energy

Since the plane is smooth (no friction), the total mechanical energy at the release point (height h) equals the total mechanical energy at the top of the loop (height $2R$).

- Initial Energy (at height h): $PE = mgh$, $KE = 0$
- Final Energy (at top of loop): $PE = mg(2R)$, $KE = \frac{1}{2}mv^2$

Equating the two:

$$mgh = mg(2R) + \frac{1}{2}mv^2$$

Cancel m from both sides:

$$gh = 2gR + \frac{1}{2}v^2$$

3. Solve for h

Substitute the value of v^2 from Equation 1 ($v^2 = 4gR$) into the energy equation:

$$gh = 2gR + \frac{1}{2}(4gR)$$

$$gh = 2gR + 2gR$$

$$gh = 4gR$$

$$h = 4R$$

Question: Electric field at centre of semi-circular ring of radius 10 cm is 100 V/m. Find charge on the ring if charge distribution is uniform.

Options:

- (a) $4 \epsilon_0$
- (b) $20 \epsilon_0$
- (c) $25 \epsilon_0$
- (d) $30 \epsilon_0$

Answer: (b)

Solution:

1. Identify the Given Values

- Electric Field (E): 100 V/m
- Radius (R): 10 cm = 0.1 m
- Permittivity of free space: ϵ_0

2. The Electric Field Formula

The electric field at the center of a semi-circular ring is given by:

$$E = \frac{2k\lambda}{R}$$

3. Express E in terms of Total Charge (Q)

Substitute $\lambda = \frac{Q}{\pi R}$ into the formula:

$$E = \frac{2 \cdot \left(\frac{1}{4\pi\epsilon_0}\right) \cdot \left(\frac{Q}{\pi R}\right)}{R}$$

$$E = \frac{2Q}{4\pi^2\epsilon_0 R^2}$$

$$E = \frac{Q}{2\pi^2\epsilon_0 R^2}$$

Where:

- $k = \frac{1}{4\pi\epsilon_0}$
- λ is the linear charge density ($\lambda = \frac{Q}{L}$)
- L is the length of the semi-circle ($L = \pi R$)

4. Solve for Q

Rearranging the formula to isolate Q :

$$Q = E \cdot 2\pi^2 \varepsilon_0 R^2$$

Now, plug in the numerical values:

$$Q = 100 \cdot 2\pi^2 \varepsilon_0 (0.1)^2$$

$$Q = 100 \cdot 2\pi^2 \varepsilon_0 (0.01)$$

$$Q = 2\pi^2 \varepsilon_0$$

Since $\pi^2 \approx 10$ (a common approximation used in physics competitive exams to match the provided options):

$$Q \approx 2(10)\varepsilon_0$$

$$Q = 20\varepsilon_0$$

Question: A point charge particle $Q = 3C$ is placed at point $A(0, -2, -5)$ and taken to

point $B(2, 1, 3)$ in the electric field $\vec{E} = (2x\hat{i} + 3y^2\hat{j} + 4\hat{k})$.

Find work done by electric field.

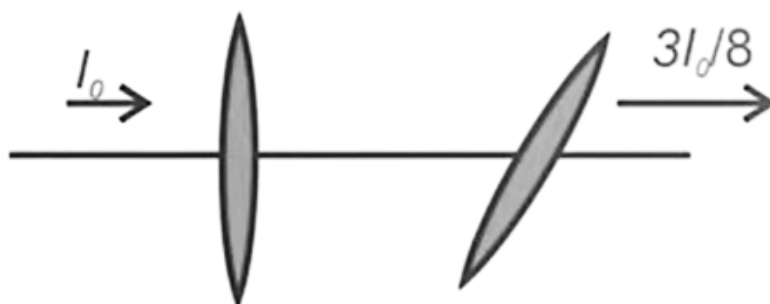
Options:

- (a) 75 Joule
- (b) 135 Joule
- (c) 89 Joule
- (d) 105 Joule

Answer: (b)

Question: Unpolarised light with intensity I_0 incident on polariser. Find angle between

axis of polariser and analyser, so that intensity of emergent light is $\frac{3I_0}{8}$.

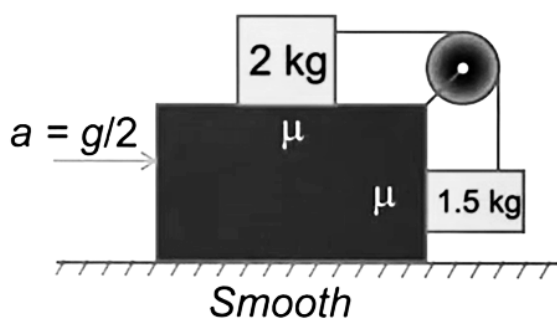


Options:

- (a) 45°
- (b) 60°
- (c) 37°
- (d) 30°

Answer: (d)

Question: Find least value of μ such that system move together without slipping.



Options:

- (a) $2/11$
- (b) $3/2$
- (c) $1/7$
- (d) $5/7$

Answer: (a)

Question: Time period of SHM of a particle is 16 sec. Find minimum time to move from

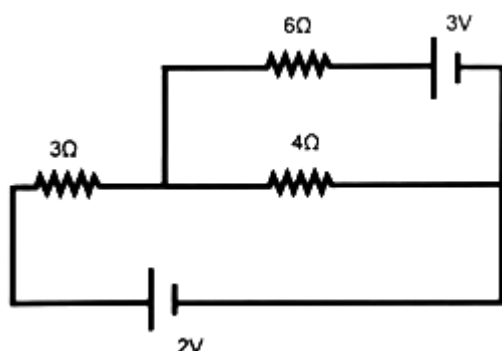
$$x = A \text{ to } x = \frac{A}{\sqrt{2}}.$$

Answer: (2 sec)

Solution:

Question: Refer to the circuit diagram given below, the heat generated across the 6Ω

resistance in 100 seconds is $\frac{\alpha}{100} J$. The value of α is _____



Answer: (3477)

Solution:

Question: Find the ratio of momentum of photons of 1st and 2nd line of balmer series of hydrogen atom.

Answer: $\frac{20}{27}$

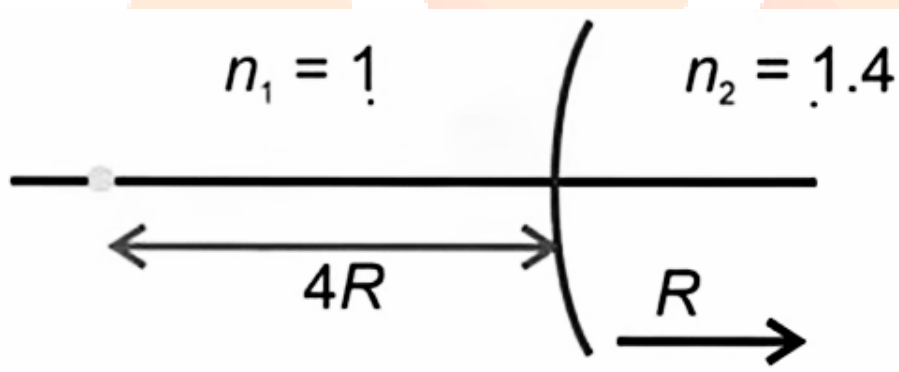
Question: A small cubical region of side 1 mm is placed at the centre of current $I = 1$ A carrying circular loop of radius $a = 1$ m. Find magnetic energy stored in the cube.

Options:

- (a) $\Delta u = \pi \times 10^{-15}$ Joule
- (b) $\Delta u = \frac{\pi}{2} \times 10^{-15}$ Joule
- (c) $\Delta u = 2\pi \times 10^{-16}$ Joule
- (d) $\Delta u = 2\pi \times 10^{-15}$ Joule

Answer: (b)

Question: A spherical refracting surface separating the media with refractive index $n_1 = 1$ and $n_2 = 1.4$ as shown. The object is placed at $4R$ from the surface and R is curvature radius as shown.

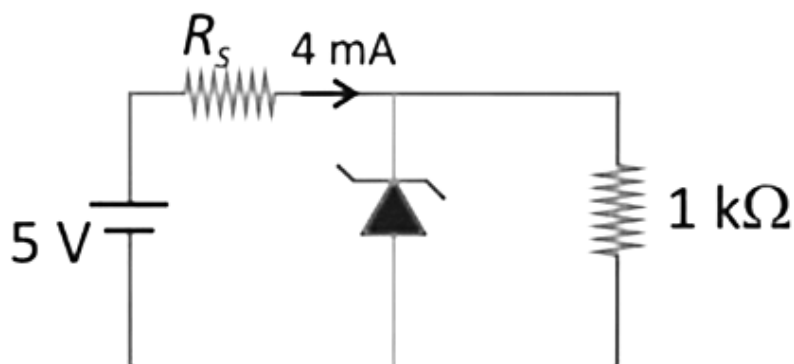


Options:

- (a) $|m| = 5/3$
- (b) $|m| = 4/3$
- (c) $|m| = 3/5$
- (d) $|m| = 3/4$

Answer: (a)

Question: Power drawn by resistance $1 \text{ k}\Omega$ is 4 mW. Find a value of R_s .



Options:

- (a) 2 kΩ
- (b) 500 Ω
- (c) 1 kΩ
- (d) 750 Ω

Answer: (d)

Question: Two wires (A & B) of same cross sectional area are subjected to equal tensile forces gets elongated by same amount. The ratio of young's modulus of two materials is $Y_A/Y_B =$ Find the ratio of lengths of the wire A?

Options:

- (a) 10/11
- (b) 20/11
- (c) 11/10
- (d) 11/20

Answer: (b)

Question: In YDSE, path difference for two points P and Q are $\frac{\lambda}{3}$ and $\frac{\lambda}{6}$ respectively. Ratio of intensities at P and Q is

Options:

- (a) 1/4
- (b) 1/3
- (c) 4
- (d) 2

Answer: (b)

Question: A one mole sample of gas undergoes an isothermal process in which volume changes from 3 litre to 1 litre. Find work done on the gas if initial pressure was 3×10^7 Pa.

Options:

- (a) 270 ln3
- (b) 900 ln3
- (c) 3000 ln3
- (d) 90000 n3

Answer: (d)

Question: Consider nuclear reaction, $P + {}_3^7\text{Li} \rightarrow 2{}_2^4\text{He}$

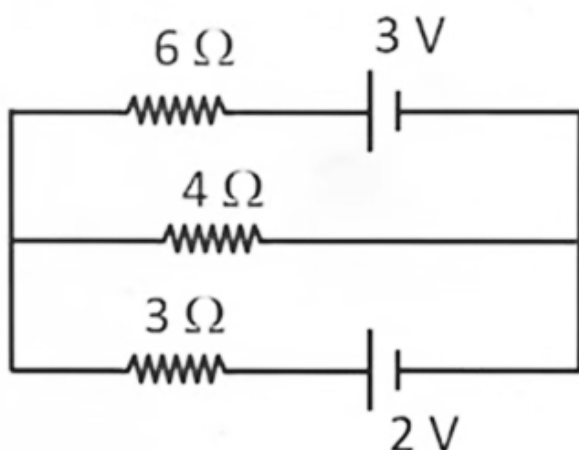
Energy absorbed in the process is ($m_p = 1.0080u$, $m_{Li} = 7.0173u$, $m_{He} = 4.0201u$)

Options:

- (a) 14 MeV
- (b) 24 MeV
- (c) 18 MeV
- (d) 11 MeV

Answer: (a)

Question: Heat dissipation in 6Ω resistance in 54 sec is $\frac{\alpha}{9} J$, then α is



Answer: (169)

Question: Mass and radius of a solid sphere is measured with uncertainty of 1% and 2% respectively uncertainty (in percent) in calculation of density is

Answer: (7)

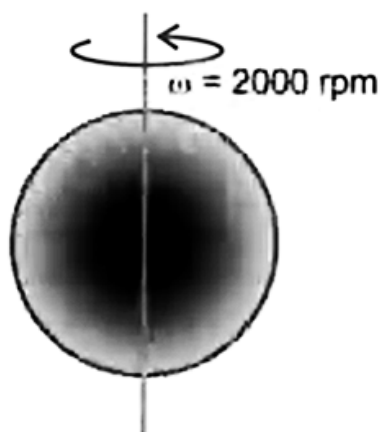
Question: The rain drop of mass $1g$, starts with zero velocity from a height of $1km$. It hits the ground with a speed of $5m/s$. The work done by the unknown resistive force is _____ J. (Take $g = 10 m/s^2$)

Options:

- (a) -8.35
- (b) -9.98
- (c) -8.75
- (d) -9.55

Answer: (b)

Question: A solid uniform sphere of mass m and radius R initially rotating with angular speed of $\omega = 2000 \text{ rpm}$. If it stops in 10 sec when subjected to some constant retarding torque, then find number of revolutions and magnitude of torque.



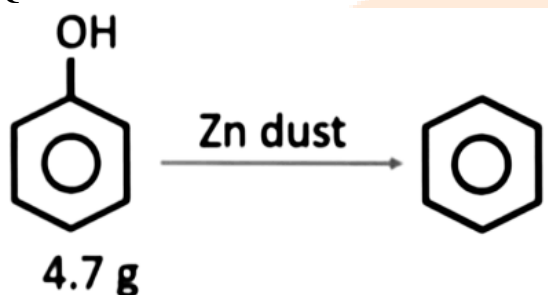
Options:

- (a) $200; \frac{8\pi}{3}mR^2$
- (b) $500; \frac{10\pi}{3}mR^2$
- (c) $\frac{500}{3}; \frac{8\pi}{3}mR^2$
- (d) $400; \frac{10\pi}{3}mR^2$

Answer: (c)

JEE-Main-06-04-2026 (Memory Based)
[MORNING SHIFT]
Chemistry

Question: Consider the reactions:



Find the no. of moles of benzene formed (if efficiency of reaction is 60%)

Options:

- (a) 0.05
- (b) 0.06
- (c) 0.03
- (d) 0.04

Answer: (c)

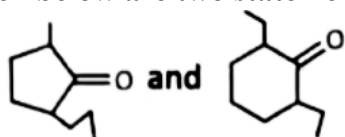
Solution: (c) 0.03

Question: Sucrose hydrolyses in acidic medium to form glucose and fructose which follows first order kinetics. If the half-life of sucrose is 3 hrs, the % of sucrose left after 6 hrs is

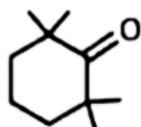
Options:

Answer: (25)

Question: Given below are two statements.



Statement-I: are metamer



Statement-II: will show tautomerism

In light of above statements, choose correct option.

Options:

- (a) Both Statement I and Statement II are correct
 (b) Statement I is incorrect, Statement II is correct
 (c) Statement I is correct, Statement II is incorrect
 (d) Both Statement I and Statement II are incorrect

Answer: (d)

Solution: (d) Both Statement I and Statement II are incorrect

Question: Match the following

List-I	List-II
P) Vitamin - A	I) Scurvy
Q) Vitamin - C	II) Convulsion
R) Vitamin - B ₆	III) Pernicious anaemia
S) Vitamin - B ₁₂	IV) Xerophthalmia

Options:

- (a) P-I, Q-III, R-II, S-IV
 (b) P-II, Q-III, R-I, S-IV
 (c) P-IV, Q-I, R-II, S-III
 (d) P-IV, Q-I, R-III, S-II

Answer: (c)

Solution: (c) P-IV, Q-I, R-II, S-III

Question: What will be the empirical formula of compound, if compound has 69.9% Fe and rest is oxygen?

Options:

- (a) Fe₃O₄
 (b) Fe₂O₃
 (c) FeO
 (d) FeO₂

Answer: (b)

Solution: (b) Fe₂O₃

Question: Statement I: 3-phenyl prop-1-ene will react with HBr and give an alkyl halide major product having 1 chiral C-atom.

Statement II: Aryl chloride and aryl cyanide both can be formed by Gattermann and Sandmeyer reaction.

Options:

- (a) Statement I is correct, Statement II is incorrect
 (b) Statement I is incorrect, Statement II is correct
 (c) Both Statement I and Statement II are correct
 (d) Both Statement I and Statement II are incorrect

Answer: (d)

Solution: (d) Both Statement I and Statement II are incorrect

Question: If the wavelength of first line of Lyman Series for H-atom is $x \text{ \AA}$. Then the longest wavelength in Balmer series of He^+ ion will be

Options:

- (a) $\frac{5x}{9}$
 (b) $\frac{27x}{20}$
 (c) $\frac{x}{27}$
 (d) $\frac{5x}{27}$

Answer: (b)

Solution: (b) $\frac{27x}{20}$

Question: Which of the following can be used to form CH_4 gas?

Options:

- (a) Kolbe's electrolysis
 (b) From CH_3MgBr
 (c) By Wurtz reaction
 (d) By reduction of $\text{CH}_3\text{CH}_2\text{Cl}$

Answer: (b)

Solution: (b) From CH_3MgBr

Question: Match List-I with List-II and choose the correct option.

List-I	List-II
A) 2s	I) One radial node + zero angular node
B) 3s	II) Two radial node + zero angular node
C) 3p	III) One radial node + 1 angular node
D) 3d	IV) Zero radial node + 2 angular node

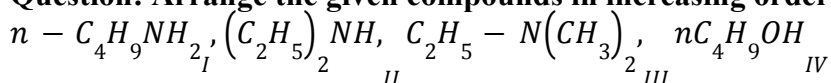
Options:

- (a) A-I, B-II, C-III, D-IV
 (b) A-II, B-IV, C-III, D-I
 (c) A-II, B-I, C-III, D-IV
 (d) A-IV, B-III, C-II, D-I

Answer: (a)

Solution: (a) A-I, B-II, C-III, D-IV

Question: Arrange the given compounds in increasing order of BP



Options:

- (a) II < III < I < IV
 (b) I < II < IV < III

(c) $\text{III} < \text{II} < \text{I} < \text{IV}$

(d) $\text{III} < \text{I} < \text{II} < \text{IV}$

Answer: (c)

Solution: (c) $\text{III} < \text{II} < \text{I} < \text{IV}$

Question: 0.25 moles of a non volatile and non ionisable compound is mixed with 1 mole of solvent. Find the percentage relative lowering in vapour pressure

Options:

(a) 20%

(b) 40%

(c) 60%

(d) 35%

Answer: (a)

Solution: (a) 20%

Question: Find the limiting molar conductivity of BaSO_4

[Given : $\Lambda_m^\circ(\text{HCl}) = x_1$

$\Lambda_m^\circ(\text{H}_2\text{SO}_4) = x_2$

$\Lambda_m^\circ(\text{BaCl}_2) = x_3]$

Options:

(a) $x_2 + x_3 - 2x_1$

(b) $x_2 + x_3 - 2x_1$

(c) $x_2 + x_3 - x_1$

(d) $x_2 + x_3 + 2x_1$

Answer: (a)

Solution: (a) $x_2 + x_3 - 2x_1$

Question: Match compounds in column-I with Column-II

Column-I	Column-II
a) Lysine	i) Hinsberg Test
b) Glutamine	ii) Hoffmann Bromamide (starting material)
c) Serine	iii) Ceric ammonium sulphate
d) Tyrosine	iv) Neutral FeCl_3 test

Options:

(a) a-i, b-ii, c-iii, d-iv

(b) a-ii, b-i, c-iii, d-iv

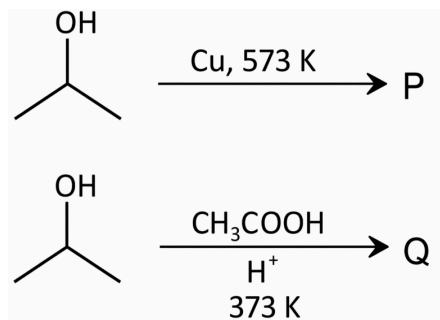
(c) a-i, b-ii, c-iv, d-iii

(d) a-ii, b-i, c-iv, d-iii

Answer: (a)

Solution: (a) a-i, b-ii, c-iii, d-iv

Question:



Compound P and Q are respectively

Options:

- (a) CH_3COCH_3 , $(\text{CH}_3)_2\text{CHOOCCCH}_3$
- (b) $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3$, $\text{CH}_3\text{-CH}=\text{CH}_2$
- (c) $\text{C}_2\text{H}_5\text{CHO}$, $\text{CH}_3\text{COOC}_2\text{H}_5$
- (d) $\text{CH}_3\text{CH}_2\text{CH}_3$, $\text{CH}_3\text{COOC}_3\text{H}_7$

Answer: (a)

Solution: (a) CH_3COCH_3 , $(\text{CH}_3)_2\text{CHOOCCCH}_3$

Question: The correct order of magnitude of work done in the following cases for some volume change and pressure change as mentioned:

Case I (a): Expansion in single step ($P_1 \rightarrow P_2$)

Case II (b): Expansion in multistep ($P_1 \rightarrow P_2$)

Case III (c): Compression in single step ($P_2 \rightarrow P_1$)

Case IV (d): Compression in multistep ($P_2 \rightarrow P_1$)

Options:

- (a) $W_C > W_D > W_B > W_A$
- (b) $W_D > W_C > W_B > W_A$
- (c) $W_A > W_B > W_D > W_C$
- (d) $W_A > W_B > W_C > W_D$

Answer: (a)

Solution: (a) $W_C > W_D > W_B > W_A$

Question: 2-Methyl propene is added into a solvent (mixture of ethyl acetate and ether) and the R_f value is found to be 0.42.

If the same 2-methyl propene is reacted with dilute H_2SO_4 , the major product obtained is x. Find the value of x.

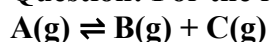
Options:

- (a) 0.12
- (b) 0.42
- (c) 0.65
- (d) 0.84

Answer: (a)

Solution: (a) 0.12

Question: For the reaction,



Consider the data

log K_p	3.5	2.5
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$\frac{1}{T}(K^{-1})$	0.04	0.05
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Calculate the value of $\frac{\Delta H}{R}$ (in K) based on above data.

Options:

Answer: (230)

Question: Choose the correct option of order of energy for square planar complex

Options:

(a) $dxz = dyz < dx^2 - y^2$

(b) $dxy = dxz > dz^2$

(c) $dxy = dyz = dxz$

(d) $dxy < dz^2 < dx^2 - y^2$

Answer: (a)

Solution: (a) $dxz = dyz < dx^2 - y^2$

Question: Consider the Levis dot structure; in following pair.

A) SO_3^{2-} and CO_3^{2-}

B) O_2 and F_2

C) NH_3 and H_3O^+

D) CN^- and CO

Options:

(a) A and C Only

(b) A, C, D Only

(c) B, C & D Only

(d) C & D Only

Answer: (d)

Solution: (d) C & D Only

JEE-Main-06-04-2026 (Memory Based)
[MORNING SHIFT]
Maths

Question: The value of $13 - 23 + 33 - 43 + \dots + 153$ is equal to

Options:

- (a) 1856
- (b) 1854
- (c) 1786
- (d) 1826

Answer: (a)

Solution:

3. Calculate the Sum of Even Cubes

The even terms are $2^3 + 4^3 + 6^3 + \dots + 14^3$. We can factor out 2^3 (which is 8):

$$8(1^3 + 2^3 + 3^3 + \dots + 7^3)$$

Using the formula again for $n = 7$:

$$\text{Even Sum} = 8 \times \left[\frac{7(8)}{2} \right]^2 = 8 \times (28)^2 = 8 \times 784 = 6,272$$

4. Final Calculation

Substitute these values back into our original rewritten expression:

$$S = 14,400 - 2(6,272)$$

$$S = 14,400 - 12,544$$

$$S = 1856$$

3. Calculate the Sum of Even Cubes

The even terms are $2^3 + 4^3 + 6^3 + \dots + 14^3$. We can factor out 2^3 (which is 8):

$$8(1^3 + 2^3 + 3^3 + \dots + 7^3)$$

Using the formula again for $n = 7$:

$$\text{Even Sum} = 8 \times \left[\frac{7(8)}{2} \right]^2 = 8 \times (28)^2 = 8 \times 784 = 6,272$$

4. Final Calculation

Substitute these values back into our original rewritten expression:

$$S = 14,400 - 2(6,272)$$

$$S = 14,400 - 12,544$$

$$S = 1856$$

Question: $\int_{-\pi/4}^{\pi/4} \frac{32\cos^4\theta}{1 + e^{\sin\theta}} d\theta$ is equal to

Options:

- (a) $3\pi + 8$
- (b) $3\pi + 4$
- (c) $4\pi + 8$

(d) $4\pi + 4$

Answer: (a)

Solution:

1. Apply the Integral Property

A useful property for integrals with symmetric limits is $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$.
In this case, the limits are $-\pi/4$ and $\pi/4$, so $a+b=0$. Replacing θ with $-\theta$:

$$I = \int_{-\pi/4}^{\pi/4} \frac{32 \cos^4(-\theta)}{1 + e^{\sin(-\theta)}} d\theta = \int_{-\pi/4}^{\pi/4} \frac{32 \cos^4 \theta}{1 + e^{-\sin \theta}} d\theta$$

(Since $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$)

2. Add the Integrals

Now, add the original integral and the one we just transformed:

$$2I = \int_{-\pi/4}^{\pi/4} 32 \cos^4 \theta \left(\frac{1}{1 + e^{\sin \theta}} + \frac{1}{1 + e^{-\sin \theta}} \right) d\theta$$

Notice that the term in the parentheses simplifies perfectly:

$$\frac{1}{1 + e^{\sin \theta}} + \frac{e^{\sin \theta}}{e^{\sin \theta} + 1} = \frac{1 + e^{\sin \theta}}{1 + e^{\sin \theta}} = 1$$

So the equation reduces to:

$$2I = \int_{-\pi/4}^{\pi/4} 32 \cos^4 \theta d\theta \Rightarrow I = 16 \int_{-\pi/4}^{\pi/4} \cos^4 \theta d\theta$$

3. Evaluate the Remaining Integral

Since $\cos^4 \theta$ is an even function, we can change the limits and double the value:

$$I = 32 \int_0^{\pi/4} \cos^4 \theta d\theta$$

Using the identity $\cos^4 \theta = \frac{1}{8} (3 + 4 \cos 2\theta + \cos 4\theta)$:

$$I = 32 \int_0^{\pi/4} \frac{1}{8} (3 + 4 \cos 2\theta + \cos 4\theta) d\theta$$

$$I = 4 \int_0^{\pi/4} (3 + 4 \cos 2\theta + \cos 4\theta) d\theta$$

Now, integrate term by term:

$$I = 4 \left[3\theta + \frac{4 \sin 2\theta}{2} + \frac{\sin 4\theta}{4} \right]_0^{\pi/4}$$

$$I = 4 \left[3\theta + 2 \sin 2\theta + \frac{1}{4} \sin 4\theta \right]_0^{\pi/4}$$

4. Substitute the Limits

- At $\pi/4$: $3(\frac{\pi}{4}) + 2\sin(\frac{\pi}{2}) + \frac{1}{4}\sin(\pi) = \frac{3\pi}{4} + 2 + 0$

- At 0: $0 + 0 + 0 = 0$

Final calculation:

$$I = 4 \left(\frac{3\pi}{4} + 2 \right) = 3\pi + 8$$

Question: The value of $\lim_{x \rightarrow 0} \frac{x^2 \sin^2 x}{x^2 - \sin^2 x}$ is equal to

Solution:

1. Identify the Indeterminate Form

As $x \rightarrow 0$:

- Numerator: $x^2 \sin^2 x \rightarrow 0 \cdot 0 = 0$
- Denominator: $x^2 - \sin^2 x \rightarrow 0 - 0 = 0$

This is a $\frac{0}{0}$ Indeterminate form.

2. Simplify the Expression

We know the standard limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. Let's rewrite the expression to isolate this:

$$\frac{x^2 \sin^2 x}{x^2 - \sin^2 x} = \left(\frac{\sin x}{x} \right)^2 \cdot \frac{x^4}{x^2 - \sin^2 x}$$

Since the left part approaches $1^2 = 1$, we only need to focus on evaluating:

$$\lim_{x \rightarrow 0} \frac{x^4}{x^2 - \sin^2 x}$$

3. Use Taylor Series Expansion

Recall the Taylor expansion for $\sin x$:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

Squaring this expansion (keeping only the lowest-order terms):

$$\sin^2 x = \left(x - \frac{x^3}{6} + \dots \right)^2$$

$$\sin^2 x = x^2 - 2(x)\left(\frac{x^3}{6}\right) + \dots = x^2 - \frac{x^4}{3} + \dots$$

Now, substitute this back into our denominator:

$$x^2 - \sin^2 x = x^2 - \left(x^2 - \frac{x^4}{3} + \dots \right) = \frac{x^4}{3} + \dots$$

4. Final Calculation

Substitute the simplified denominator back into the limit:

$$\lim_{x \rightarrow 0} \frac{x^4}{\frac{x^4}{3} + \text{higher order terms}}$$

Cancel the x^4 terms:

$$\lim_{x \rightarrow 0} \frac{1}{\frac{1}{3}} = 3$$

Question: If $x_1, x_2, \dots, x_{25}$ be 25 observations such that Then, the ratio of Mean and Standard deviation of the given observation is

Options:

- (a) $\frac{1}{4}$

(b) $\frac{1}{2}$

(c) $\frac{2}{3}$

(d) $\frac{1}{5}$

Answer: (b)

Solution:

1. Expand the Given Equations

We are given two equations for $n = 25$ observations:

$$1. \sum (x_i + 5)^2 = 2500$$

$$2. \sum (x_i - 5)^2 = 1000$$

Expanding equation (1):

$$\sum (x_i^2 + 10x_i + 25) = 2500$$

$$\sum x_i^2 + 10 \sum x_i + (25 \times 25) = 2500$$

$$\sum x_i^2 + 10 \sum x_i = 1875 \quad \text{---(Eq. A)}$$

Expanding equation (2):

$$\sum (x_i^2 - 10x_i + 25) = 1000$$

$$\sum x_i^2 - 10 \sum x_i + (25 \times 25) = 1000$$

$$\sum x_i^2 - 10 \sum x_i = 375 \quad \text{---(Eq. B)}$$

2. Solve for the Sums

Subtract (Eq. B) from (Eq. A) to find the sum of observations:

$$(10 \sum x_i) - (-10 \sum x_i) = 1875 - 375$$

$$20 \sum x_i = 1500 \implies \sum x_i = 75$$

Add (Eq. A) and (Eq. B) to find the sum of squares:

$$2 \sum x_i^2 = 1875 + 375$$

$$2 \sum x_i^2 = 2250 \implies \sum x_i^2 = 1125$$

3. Calculate Mean ($\bar{x}$) and Standard Deviation (σ)

- **Mean:**

$$\bar{x} = \frac{\sum x_i}{n} = \frac{75}{25} = 3$$

- **Standard Deviation:**

First, find the variance (σ^2):

$$\sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$\sigma^2 = \frac{1125}{25} - (3)^2 = 45 - 9 = 36$$

$$\sigma = \sqrt{36} = 6$$

4. Find the Ratio

The question asks for the ratio of the **Mean** to the **Standard Deviation**:

$$\text{Ratio} = \frac{\bar{x}}{\sigma} = \frac{3}{6} = \frac{1}{2}$$

Question: In the expansion of $(1 + ax)^{26}$ and $(1 - ax)^{28}$, the coefficient of middle term is same, then the value of a is

Options:

- (a) $\frac{7}{22}$
- (b) $\frac{7}{27}$
- (c) $\frac{5}{27}$
- (d) $\frac{5}{22}$

Answer: (b)

Solution:

1. Middle Term of $(1 + ax)^{26}$

For a binomial expansion $(x + y)^n$ where n is even, the middle term is the $(\frac{n}{2} + 1)^{\text{th}}$ term.

- Here, $n = 26$, so the middle term is the $(\frac{26}{2} + 1) = 14^{\text{th}}$ term.
- Using the general term formula $T_{r+1} = {}^nC_r \cdot x^{n-r} \cdot y^r$, for the 14^{th} term ($r = 13$):

$$T_{14} = {}^{26}C_{13} \cdot (ax)^{13}$$

- **Coefficient** = ${}^{26}C_{13} \cdot a^{13}$

2. Middle Term of $(1 - ax)^{28}$

- Here, $n = 28$, so the middle term is the $(\frac{28}{2} + 1) = 15^{\text{th}}$ term.

- For the 15th term ($r = 14$):

$$T_{15} = {}^{28}C_{14} \cdot (-ax)^{14} = {}^{28}C_{14} \cdot a^{14} \cdot x^{14}$$

(Note: $(-1)^{14} = 1$ because the power is even)

- Coefficient = ${}^{28}C_{14} \cdot a^{14}$

3. Equate the Coefficients and Solve for a

According to the problem, these coefficients are the same:

$${}^{26}C_{13} \cdot a^{13} = {}^{28}C_{14} \cdot a^{14}$$

Divide both sides by a^{13} :

$$a = \frac{{}^{26}C_{13}}{{}^{28}C_{14}}$$

Now, expand the combinations:

$$a = \frac{\frac{26!}{13!13!}}{\frac{28!}{14!14!}}$$

$$a = \frac{26!}{13!13!} \times \frac{14!14!}{28!}$$

Simplify by pairing terms:

$$a = \left(\frac{26!}{28!} \right) \times \left(\frac{14!}{13!} \right) \times \left(\frac{14!}{13!} \right)$$

$$a = \left(\frac{1}{28 \times 27} \right) \times 14 \times 14$$

$$a = \frac{196}{756}$$

Reducing the fraction:

$$a = \frac{14 \times 14}{28 \times 27} = \frac{1 \times 14}{2 \times 27} = \frac{7}{27}$$

Question: $a_1, a_2, a_3, \dots, a_n$ are in A.P. and sum of first 10 terms is 160. $g_1, g_2, g_3, \dots, g_n$ are in G.P., where $g_1 + g_2 = 8$. If the first term of A.P. is equal to common ratio of G.P. and first term of G.P. is equal to common difference of A.P., then sum of all possible values of g_1 is equal to

Options:

- (a) $\frac{34}{9}$
- (b) $\frac{28}{9}$
- (c) $\frac{23}{3}$
- (d) $\frac{28}{5}$

Answer: (a)

Solution:

1. Identify Relations from the A.P.

For the A.P. $a_1, a_2, \dots, a_n$, let a_1 be the first term and d be the common difference. The sum of the first 10 terms (S_{10}) is given by:

$$S_{10} = \frac{n}{2} [2a_1 + (n-1)d]$$

$$160 = \frac{10}{2} [2a_1 + 9d]$$

$$160 = 5[2a_1 + 9d] \Rightarrow 2a_1 + 9d = 32 \quad \text{---(Eq. 1)}$$

2. Identify Relations from the G.P.

For the G.P. $g_1, g_2, \dots, g_n$, let g_1 be the first term and r be the common ratio. We are given:

$$g_1 + g_2 = 8$$

$$g_1 + g_1 r = 8 \Rightarrow g_1(1 + r) = 8 \quad \text{---(Eq. 2)}$$

3. Apply the Interlinking Conditions

The problem gives us two specific links between the progressions:

1. First term of A.P. = Common ratio of G.P.: $a_1 = r$

2. First term of G.P. = Common difference of A.P.: $g_1 = d$

Substituting these into Eq. 1, we can express r in terms of g_1 :

$$2r + 9g_1 = 32 \Rightarrow r = \frac{32 - 9g_1}{2}$$

Now, substitute this expression for r into Eq. 2:

$$g_1 \left(1 + \frac{32 - 9g_1}{2} \right) = 8$$

$$g_1 + g_2 = 8$$

$$g_1 + g_1 r = 8 \implies g_1(1 + r) = 8 \quad \text{---(Eq. 2)}$$

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1. First term of A.P. = Common ratio of G.P.: $a_1 = r$

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Substituting these into Eq. 1, we can express r in terms of g_1 :

$$2r + 9g_1 = 32 \implies r = \frac{32 - 9g_1}{2}$$

Now, substitute this expression for r into Eq. 2:

$$g_1 \left(1 + \frac{32 - 9g_1}{2} \right) = 8$$

$$g_1 \left(\frac{2 + 32 - 9g_1}{2} \right) = 8$$

$$g_1(34 - 9g_1) = 16$$

$$-9g_1^2 + 34g_1 - 16 = 0 \implies 9g_1^2 - 34g_1 + 16 = 0$$

4. Calculate the Sum of Possible Values

The possible values of g_1 are the roots of the quadratic equation $9g_1^2 - 34g_1 + 16 = 0$.

For any quadratic equation $Ax^2 + Bx + C = 0$, the sum of the roots is given by $-\frac{B}{A}$.

$$\text{Sum of } g_1 \text{ values} = -\frac{(-34)}{9} = \frac{34}{9}$$

Question: Consider e_1 and e_2 be roots of the equation $x^2 - ax + 2 = 0$. Set of values of a for which e_1 and e_2 are eccentricities of hyperbolas then $a \in [\alpha, \beta]$ and set of values of a for which e_1 and e_2 are eccentricity of a hyperbola and an ellipse is (γ, ∞) then $\alpha^2 + \beta^2 + \gamma^2$ is equal to

Options:

- (a) 26
- (b) 24
- (c) 18
- (d) 32

Answer: (a)

Solution:

1. Identify the Properties of the Roots

For the equation $x^2 - ax + 2 = 0$, let the roots be e_1 and e_2 . From the properties of quadratic equations:

- **Sum of roots:** $e_1 + e_2 = a$
- **Product of roots:** $e_1 e_2 = 2$
- **Discriminant (D):** For the roots to be real, $D = a^2 - 4(1)(2) = a^2 - 8 \geq 0$. This means $a \geq \sqrt{8}$ or $a \leq -\sqrt{8}$. Since eccentricities are positive, we have $a \geq 2\sqrt{2}$.

2. Case 1: Both e_1 and e_2 are eccentricities of hyperbolas

For a hyperbola, the eccentricity must be greater than 1 ($e > 1$). For both roots to be greater than 1:

$$1. D \geq 0: a^2 - 8 \geq 0 \implies a \geq 2\sqrt{2}$$

$$2. f(1) > 0: \text{Let } f(x) = x^2 - ax + 2. \text{ If both roots are } > 1, \text{ then } f(1) = 1 - a + 2 = 3 - a > 0, \text{ which means } a < 3.$$

$$3. \text{Vertex Location: The vertex of the parabola } (x = a/2) \text{ must be greater than 1, so } a/2 > 1 \implies a > 2.$$

Combining these, we get $a \in [2\sqrt{2}, 3)$. In the context of the problem's notation $a \in [\alpha, \beta]$, the boundary values are:

- $\alpha = 2\sqrt{2}$
- $\beta = 3$

3. Case 2: One root is an ellipse and one is a hyperbola

For an ellipse, $0 < e < 1$. For a hyperbola, $e > 1$.

Since the product $e_1 e_2 = 2$, if one root is in the interval $(0, 1)$, the other root must be $2/(something < 1)$, which is automatically greater than 2 (and thus a hyperbola).

To have roots on either side of $x = 1$, we must have $f(1) < 0$:

$$1 - a + 2 < 0 \implies 3 - a < 0 \implies a > 3$$

This gives the interval $(3, \infty)$. Comparing this to (γ, ∞) :

- $\gamma = 3$

4. Final Calculation

Now, we calculate the required value $\alpha^2 + \beta^2 + \gamma^2$:

- $\alpha^2 = (2\sqrt{2})^2 = 8$
- $\beta^2 = 3^2 = 9$
- $\gamma^2 = 3^2 = 9$

$$\text{Total} = 8 + 9 + 9 = 26$$

Question: There are $(n + 1)$ coins. 'N' coins are unbiased coins and one coin has two heads. A coin is randomly chosen and tossed once. If the probability of getting head is $\frac{9}{16}$, then the value of n is

Options:

- (a) 5
- (b) 6
- (c) 7
- (d) 8

Answer: (c)

Solution:

1. Identify the Possible Events

There are a total of $(n + 1)$ coins in the bag.

- **Event U (Choosing an unbiased coin):** There are n such coins.

$$P(U) = \frac{n}{n+1}$$
- **Event B (Choosing the biased/two-headed coin):** There is 1 such coin.

$$P(B) = \frac{1}{n+1}$$

2. Determine the Probability of a Head (H) for each coin

- For an unbiased coin, $P(H|U) = \frac{1}{2}$.
- For a two-headed coin, $P(H|B) = 1$ (it always shows a head).

3. Set Up the Total Probability Equation

The total probability of getting a head is the sum of the probabilities from both cases:

$$P(H) = [P(U) \cdot P(H|U)] + [P(B) \cdot P(H|B)]$$

Substitute the values we found:

$$\frac{9}{16} = \left(\frac{n}{n+1} \cdot \frac{1}{2} \right) + \left(\frac{1}{n+1} \cdot 1 \right)$$

4. Solve for n

Combine the terms over a common denominator:

$$\frac{9}{16} = \frac{n}{2(n+1)} + \frac{2}{2(n+1)}$$

$$\frac{9}{16} = \frac{n+2}{2(n+1)}$$

Cross-multiply to solve the equation:

$$9 \cdot 2(n+1) = 16(n+2)$$

$$18(n+1) = 16n + 32$$

$$18n + 18 = 16n + 32$$

Now, isolate n :

$$18n - 16n = 32 - 18$$

$$2n = 14$$

$$n = 7$$

Question: The number of 4 letters words which can be made using the letters of the word INCONSEQUENTIAL without repetition using 2 vowels and 2 consonants is equal to

Options:

- (a) 3460
- (b) 3600
- (c) 4200
- (d) 2400

Answer: (b)

Solution:

1. Analyze the word: INCONSEQUINTAL

First, let's list the frequency of each letter in the word:

- **Vowels:** I (2), O (1), E (1), U (1), A (1)
 - *Distinct vowels:* {I, O, E, U, A} — Total 5 types.
- **Consonants:** N (3), C (1), S (1), Q (1), T (1), L (1)
 - *Distinct consonants:* {N, C, S, Q, T, L} — Total 6 types.

The phrase "without repetition" means that the 4 letters chosen for the word must be distinct from one another. Therefore, we pick from the distinct types available.

2. Selection (Combination)

We need to select 2 vowels from the 5 distinct types and 2 consonants from the 6 distinct types:

- Ways to select 2 vowels: ${}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10$
- Ways to select 2 consonants: ${}^6C_2 = \frac{6 \times 5}{2 \times 1} = 15$

Total number of ways to select the 4 distinct letters:

$$10 \times 15 = 150$$

3. Arrangement (Permutation)

Once we have selected 4 distinct letters, they can be arranged in 4! ways to form different words:

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

4. Final Calculation

The total number of words is the product of the number of selections and the number of arrangements:

$$\text{Total Words} = 150 \times 24 = 3600$$

Question: If $\tan^{-1}(1 - \alpha) + \tan^{-1}(1 - \beta) = \frac{\pi}{4}$ and $b = \frac{1}{3\alpha}$ then the value of $6(\alpha + \beta)$ is equal to

Options:

- (a) 7
- (b) 9
- (c) 8
- (d) 6

Answer: (a)

Solution:

1. Apply the Inverse Tangent Identity

Recall the identity: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$.

Setting $x = 1 - \alpha$ and $y = 1 - \beta$, the equation becomes:

$$\tan^{-1} \left(\frac{(1-\alpha) + (1-\beta)}{1 - (1-\alpha)(1-\beta)} \right) = \frac{\pi}{4}$$

Taking the tangent of both sides ($\tan(\pi/4) = 1$):

$$\frac{2 - (\alpha + \beta)}{1 - (1 - \alpha - \beta + \alpha\beta)} = 1$$

2. Simplify the Algebra

Expand the denominator and simplify:

$$\frac{2 - (\alpha + \beta)}{1 - 1 + \alpha + \beta - \alpha\beta} = 1$$

$$\frac{2 - (\alpha + \beta)}{\alpha + \beta - \alpha\beta} = 1$$

Cross-multiply to solve:

$$2 - (\alpha + \beta) = (\alpha + \beta) - \alpha\beta$$

$$2 + \alpha\beta = 2(\alpha + \beta)$$

3. Use the Relationship $\beta = \frac{1}{3\alpha}$

We are given that $\beta = \frac{1}{3\alpha}$, which implies that their product is constant:

$$\alpha\beta = \frac{1}{3}$$

Substitute this value into our simplified equation:

$$2 + \frac{1}{3} = 2(\alpha + \beta)$$

$$\frac{7}{3} = 2(\alpha + \beta)$$

4. Final Calculation

The question asks for the value of $6(\alpha + \beta)$. Multiplying our result by 3 on both sides:

$$3 \times \left[2(\alpha + \beta) \right] = 3 \times \frac{7}{3}$$

$$6(\alpha + \beta) = 7$$

Question: If $1 + \cos x = \sqrt{3} \sin x$ where $x \in (-2\pi, 2\pi)$. Then, the sum of all the values of x satisfy the given equation is

Options:

- (a) 5π
- (b) $\frac{4\pi}{3}$
- (c) 4π
- (d) $\frac{-4\pi}{3}$

Answer: (d)

Solution:

1. Simplify the Equation

First, rearrange the equation into a standard trigonometric form:

$$\sqrt{3} \sin x - \cos x = 1$$

Divide the entire equation by 2 (since $\sqrt{(\sqrt{3})^2 + (-1)^2} = 2$):

$$\frac{\sqrt{3}}{2} \sin x - \frac{1}{2} \cos x = \frac{1}{2}$$

Using the identity $\sin A \cos B - \cos A \sin B = \sin(A - B)$, we can rewrite this as:

$$\sin x \cos\left(\frac{\pi}{6}\right) - \cos x \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\sin\left(x - \frac{\pi}{6}\right) = \frac{1}{2}$$

2. Find Solutions in the Given Range

The basic solutions for $\sin \theta = 1/2$ are $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$.

We need to find all values of x such that $\left(x - \frac{\pi}{6}\right)$ falls within the transformed interval.

Since $x \in (-2\pi, 2\pi)$, then $\left(x - \frac{\pi}{6}\right) \in \left(-\frac{13\pi}{6}, \frac{11\pi}{6}\right)$.

Let's find the values of $\theta = x - \frac{\pi}{6}$ in this interval:

$$1. \theta_1 = \frac{\pi}{6} \implies x_1 = \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$$

$$2. \theta_2 = \frac{5\pi}{6} \implies x_2 = \frac{5\pi}{6} + \frac{\pi}{6} = \pi$$

$$3. \theta_3 = \frac{\pi}{6} - 2\pi = -\frac{11\pi}{6} \implies x_3 = -\frac{11\pi}{6} + \frac{\pi}{6} = -\frac{5\pi}{3}$$

$$4. \theta_4 = \frac{5\pi}{6} - 2\pi = -\frac{7\pi}{6} \implies x_4 = -\frac{7\pi}{6} + \frac{\pi}{6} = -\pi$$

All these values of x ($-\frac{5\pi}{3}$, $-\pi$, $\frac{\pi}{3}$, π) are within the range $(-2\pi, 2\pi)$.

3. Calculate the Sum

The sum of all values of x is:

$$\text{Sum} = \left(-\frac{5\pi}{3}\right) + (-\pi) + \left(\frac{\pi}{3}\right) + \pi$$

$$\text{Sum} = -\frac{5\pi}{3} + \frac{\pi}{3}$$

$$\text{Sum} = -\frac{4\pi}{3}$$

Question: Points P and Q lie on the parabola $y^2 = 12x$. The ratio of their y-coordinates is 1 : 2 and the length of the line segment PQ is $3\sqrt{13}$. If line PQ makes an angle θ with positive x-axis in anticlockwise direction, $\theta \in (0, \pi)$ then $\sin \theta$ is equal to

Options:

- (a) $\frac{1}{\sqrt{13}}$
- (b) $\frac{\sqrt{13}}{3}$
- (c) $\frac{\sqrt{13}}{2}$
- (d) $\frac{1}{\sqrt{12}}$

Answer: (c)

Solution:

1. Identify the coordinates of P and Q

The equation of the parabola is $y^2 = 12x$. Any point (x, y) on this parabola can be written in terms of its y-coordinate as $\left(\frac{y^2}{12}, y\right)$.

Let the y-coordinate of point P be y_1 . Since the ratio of the y-coordinates is 1 : 2, the y-coordinate of point Q is $2y_1$.

- $P = \left(\frac{y_1^2}{12}, y_1\right)$

- $Q = \left(\frac{(2y_1)^2}{12}, 2y_1\right) = \left(\frac{4y_1^2}{12}, 2y_1\right) = \left(\frac{y_1^2}{3}, 2y_1\right)$

2. Use the Distance Formula

We are given that the length of the line segment PQ is $3\sqrt{13}$. Using the distance formula:

$$PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$(3\sqrt{13})^2 = \left(\frac{y_1^2}{3} - \frac{y_1^2}{12}\right)^2 + (2y_1 - y_1)^2$$

$$117 = \left(\frac{3y_1^2}{12}\right)^2 + y_1^2 = \left(\frac{y_1^2}{4}\right)^2 + y_1^2$$

$$117 = \frac{y_1^4}{16} + y_1^2$$

Multiplying the entire equation by 16 to clear the fraction:

$$y_1^4 + 16y_1^2 - 1872 = 0$$

$$y_1^4 + 16y_1^2 - 1872 = 0$$

Let $u = y_1^2$. The quadratic equation $u^2 + 16u - 1872 = 0$ can be factored:

$$(u + 52)(u - 36) = 0$$

Since $u = y_1^2$ must be positive, we have $y_1^2 = 36$, which means $y_1 = \pm 6$.

3. Find the Slope ($\tan\theta$)

The slope of the line PQ is given by $m = \tan\theta$:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2y_1 - y_1}{\frac{y_1^2}{3} - \frac{y_1^2}{12}} = \frac{y_1}{\frac{3y_1^2}{12} - \frac{y_1^2}{4}} = \frac{y_1}{\frac{y_1^2}{4}} = \frac{4}{y_1}$$

- If $y_1 = 6$, then $\tan\theta = \frac{4}{6} = \frac{2}{3}$.
- If $y_1 = -6$, then $\tan\theta = \frac{4}{-6} = -\frac{2}{3}$.

4. Calculate $\sin\theta$

Since $\theta \in (0, \pi)$, $\sin\theta$ will be positive regardless of whether the slope is positive or negative.

If $|\tan\theta| = \frac{2}{3}$, we can imagine a right triangle where the **opposite side is 2** and the **adjacent side is 3**.

The hypotenuse is $\sqrt{2^2 + 3^2} = \sqrt{13}$.

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{2}{\sqrt{13}}$$

Question: If the area (in square units) of the region $\{(x, y) : y \leq 6 - x, y \geq 4x - 3, x \geq 0, y \geq 0\}$ is

Options:

- (a) 15
- (b) 9
- (c) 8

(d) 7

Answer: (b)

Solution:

1. Identify the Boundaries

The region is defined by the following inequalities in the first quadrant ($x \geq 0, y \geq 0$):

- **Line:** $x + y \leq 6 \implies x = 6 - y$
- **Parabola:** $y^2 \leq 4x - 3 \implies x = \frac{y^2 + 3}{4}$

2. Find the Intersection Point

To find where the line and the parabola meet, we set their expressions for x equal to each other:

$$6 - y = \frac{y^2 + 3}{4}$$

$$24 - 4y = y^2 + 3$$

$$y^2 + 4y - 21 = 0$$

Factoring the quadratic equation:

$$(y + 7)(y - 3) = 0$$

Since we are given $y \geq 0$, we take $y = 3$.

When $y = 3$, $x = 6 - 3 = 3$. So the intersection point is $(3, 3)$.

3. Set Up the Integral

It is much easier to calculate this area by integrating with respect to y (from $y = 0$ to $y = 3$), as the region is bounded on the right by the line and on the left by the parabola.

$$\text{Area} = \int_0^3 (x_{\text{line}} - x_{\text{parabola}}) dy$$

$$\text{Area} = \int_0^3 \left((6 - y) - \frac{y^2 + 3}{4} \right) dy$$

Simplify the integrand:

$$\text{Area} = \int_0^3 \left(6 - y - \frac{y^2}{4} - \frac{3}{4} \right) dy = \int_0^3 \left(\frac{21}{4} - y - \frac{y^2}{4} \right) dy$$

4. Evaluate the Integral

$$\text{Area} = \left[\frac{21}{4} y - \frac{y^2}{2} - \frac{y^3}{12} \right]_0^3$$

Substitute the upper limit ($y = 3$):

$$\text{Area} = \left(\frac{21 \cdot 3}{4} - \frac{3^2}{2} - \frac{3^3}{12} \right) - 0$$

$$\text{Area} = \frac{63}{4} - \frac{9}{2} - \frac{27}{12}$$

$$\text{Area} = \frac{63}{4} - \frac{18}{4} - \frac{9}{4} = \frac{36}{4} = 9$$

Question: The domains of the function of $\alpha^2 + \beta^2$ is equal to

Options:

- (a) 4
- (b) 5
- (c) 3
- (d) 2

Answer: (b)

Solution:

1. Understand the Domain Condition

The domain of the inverse sine function, $\sin^{-1}(u)$, is restricted to $-1 \leq u \leq 1$. Therefore, for $f(x) = \sin^{-1}\left(\frac{x + [x]}{3}\right)$, we must satisfy the inequality:

$$-1 \leq \frac{x + [x]}{3} \leq 1$$

Multiplying the entire inequality by 3, we get:

$$-3 \leq x + [x] \leq 3$$

2. Solve the Inequality $x + [x] \geq -3$

Let's analyze the behavior of $g(x) = x + [x]$ for different intervals:

- If $x \in [-2, -1)$, then $[x] = -2$. The expression becomes $x - 2 \geq -3 \implies x \geq -1$. Since our interval was $x < -1$, no values in this range satisfy the condition.

$$f(x) = \sin^{-1}\left(\frac{x + [x]}{3}\right) \text{ is } [\alpha, \beta]. \text{ The value}$$

- If $x = -1$, then $[x] = -1$. The expression is $-1 + (-1) = -2$, which is ≥ -3 . This works.
 - For any $x > -1$, $x + [x]$ will be greater than -2 , which satisfies the lower bound.
- So, the minimum value for x is $\alpha = -1$.

3. Solve the Inequality $x + [x] \leq 3$

Now, let's find the upper bound:

- If $x \in [1, 2)$, then $[x] = 1$. The expression becomes $x + 1 \leq 3 \implies x \leq 2$. All values in the interval $[1, 2)$ satisfy this.
- If $x = 2$, then $[x] = 2$. The expression is $2 + 2 = 4$, which is **greater than 3**. Thus, $x = 2$ is excluded from the domain.
- For any $x > 2$, $x + [x]$ will be even larger than 4.

So, the values of x must be strictly less than 2. The maximum value for x is $\beta = 2$.

4. Identify α and β and Calculate

Combining the results, the domain is $x \in [-1, 2)$.

Comparing this to the given interval $[\alpha, \beta)$:

- $\alpha = -1$
- $\beta = 2$

Finally, we calculate the required value:

$$\alpha^2 + \beta^2 = (-1)^2 + (2)^2 = 1 + 4 = 5$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, a, b > 0$$

Question: If the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a, b > 0$ with eccentricity e passes through point $P(6, 4\sqrt{3})$ and $15(e^2 + 1) = (34)e$ then length of latus rectum of hyperbola

$$\frac{x^2}{b^2} - \frac{y^2}{2(a^2 + 1)} = 1 \text{ is}$$

Options:

- (a) 10
- (b) 12
- (c) 14
- (d) 18

Answer: (a)

Solution:

1. Solve for the eccentricity (e)

The given equation for eccentricity is $15(e^2 + 1) = 34e$. Rearranging this into a standard quadratic form:

$$15e^2 - 34e + 15 = 0$$

We can factor this quadratic equation:

$$15e^2 - 25e - 9e + 15 = 0$$

$$5e(3e - 5) - 3(3e - 5) = 0$$

$$(5e - 3)(3e - 5) = 0$$

This gives two possible values: $e = \frac{3}{5}$ or $e = \frac{5}{3}$. Since the eccentricity of a hyperbola must be greater than 1, we take:

$$e = \frac{5}{3}$$

2. Relate a^2 and b^2

For a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the relationship between the axes and eccentricity is $b^2 = a^2(e^2 - 1)$.

$$b^2 = a^2 \left(\left(\frac{5}{3} \right)^2 - 1 \right) = a^2 \left(\frac{25}{9} - 1 \right) = \frac{16a^2}{9}$$

So, $b^2 = \frac{16a^2}{9}$.

3. Use the point $P(6, 4\sqrt{3})$ to find a^2

Substitute the coordinates of point P into the hyperbola's equation:

$$\frac{6^2}{a^2} - \frac{(4\sqrt{3})^2}{b^2} = 1 \Rightarrow \frac{36}{a^2} - \frac{48}{b^2} = 1$$

Now, substitute $b^2 = \frac{16a^2}{9}$ into this equation:

$$\frac{36}{a^2} - \frac{48}{\frac{16a^2}{9}} = 1$$

$$\frac{36}{a^2} - \frac{48 \cdot 9}{16a^2} = 1 \Rightarrow \frac{36}{a^2} - \frac{27}{a^2} = 1$$

$$\frac{9}{a^2} = 1 \Rightarrow a^2 = 9$$

Then, $b^2 = \frac{16}{9} \cdot 9 = 16$.

4. Calculate the Latus Rectum of the second hyperbola

The second hyperbola equation is $\frac{x^2}{b^2} - \frac{y^2}{2(a^2 + 1)} = 1$. Substituting our values:

$$\frac{x^2}{16} - \frac{y^2}{2(9 + 1)} = 1 \Rightarrow \frac{x^2}{16} - \frac{y^2}{20} = 1$$

For this hyperbola, let $A^2 = 16$ (so $A = 4$) and $B^2 = 20$. The formula for the length of the latus rectum is:

$$L.R. = \frac{2B^2}{A}$$

$$L.R. = \frac{2(20)}{4} = \frac{40}{4} = 10$$